

EDIDIONG DAVID IBOKETE

DATA SCIENTIST | MACHINE LEARNING RESEARCHER | APPLIED AI ENGINEER

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PROFESSIONAL SUMMARY

Machine learning researcher and data scientist with about three years of experience building reliable data, machine learning, and applied AI systems across IoT cybersecurity, retrieval-augmented generation, forecasting, institutional analytics, and geospatial operations. Combines Python, SQL, statistical modeling, data engineering, cloud deployment, and rigorous evaluation to translate complex requirements into reproducible decision-support systems.

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, SAS, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, XGBoost

Machine Learning & Statistics: Regression, classification, time-series forecasting, anomaly detection, feature engineering, SHAP, hypothesis testing, model validation, explainability

Applied AI & MLOps: RAG, embeddings, LLMs, prompt engineering, FastAPI, Azure App Service, OpenAI models, knowledge-base design, retrieval evaluation, CI/CD concepts

Data Engineering & BI: ETL, Alteryx, MS SQL Server, MySQL, SLATE CRM, data validation, Tableau, Power BI, Git/GitHub, reproducible pipelines

PROFESSIONAL EXPERIENCE

Center for Research Libraries | Chicago, IL

Algorithm & Data Engineer Intern

Jun 2025 - Aug 2025

- Translated cooperative cataloging requirements across 19 institutions into a Python/Pandas capacity-constrained matching pipeline that modeled 194 tasks using priority, qualification fit, reciprocity, trust, and contributor capacity.
- Engineered and validated normalized requester and contributor profiles from survey data, converting inconsistent institutional inputs into reviewable scoring, assignment, and diagnostic outputs.
- Built an anonymized reproducibility package that generated 565 scored candidate matches, passed end-to-end smoke testing, and enabled transparent technical review without exposing private institutional data.
- Collaborated with stakeholders to translate operational workflows into implementation-ready allocation rules, validation checkpoints, and technical documentation.

Canisius University | Buffalo, NY

Graduate Assistant, Data Analytics & Reporting

Aug 2024 - Dec 2025

- Engineered automated ETL and validation workflows using SQL, SAS PROC SQL, R, and Alteryx across SLATE CRM and MS SQL Server, supporting enrollment reporting and downstream institutional analytics.
- Identified and removed duplicate and inconsistent registration records with 100% deduplication accuracy, strengthening dataset reliability and reducing downstream correction effort.
- Optimized relational queries and validation logic, reducing record redundancy by 70% and improving structured reporting reliability.
- Developed Tableau and Power BI dashboards and documented data flows, reporting logic, and validation checkpoints to support institutional planning, maintainability, and knowledge transfer.

Geospatial Intelligence and Data Analysis Centre, Ministry of Defense | Abuja, Nigeria

Pioneer Database Administrator & Geospatial Data Analyst

2021 - 2023

- Administered MySQL and MS SQL Server environments and secure access controls for structured and geospatial datasets, maintaining data integrity and reporting availability in mission-critical operations.
- Improved query efficiency by 30% through indexing strategies and query restructuring under high-demand workloads.
- Built Python-assisted analytical workflows that accelerated monthly reporting by 25% and integrated geospatial and structured data into analytics-ready pipelines for faster decision support.

SELECTED RESEARCH & APPLIED AI PROJECTS

Certified Anomaly Detection for IoT Security - Co-Author, Research Software & Evaluation | Submitted / Submitted Under Review

- Implemented and evaluated a Negative-Binomial concentration framework for overdispersed IoT traffic, achieving 1.000 AUC, 0.000 certified-mode benign FPR, and 448 microseconds latency per 10 ms window; NB fit better than Poisson on 98.5% of applicable Gotham devices.
- Developed the Poisson-CI evaluation pipeline, attaining 0.9648 ROC-AUC and 0.9708 F1 at a matched 1% FPR across eight attack families, with cross-dataset validation against classical anomaly-detection baselines.
- Modularized Python research components, created configuration-driven experiments, automated notebook execution, and audited figures, tables, and manuscript values for reproducibility and reviewer-ready release.

Project ARIEL - Deployed RAG Portfolio Assistant | Azure App Service / FastAPI / OpenAI

- Designed and deployed an end-to-end RAG application that connects a curated YAML/Markdown knowledge base, embedding retrieval, grounded generation, a FastAPI backend, and a JavaScript widget embedded across a GitHub Pages portfolio.
- Built 11 retrieval evaluation suites covering 184 audit questions; in the latest report, only three questions missed the expected document at rank 1, while every expected document remained within the top three retrieved results.

Commercial Building Energy Forecasting & Optimization | Predictive Modeling Lead | 1st Place

- Built an XGBoost forecasting pipeline with time-series, building, and weather features, achieving 0.9804 R-squared, 5.44 MAE, and 34.28 RMSE on future-period evaluation across a 1,600+ building commercial portfolio.
- Used SHAP, residual diagnostics, and savings simulations to identify 10 underperforming buildings and estimate \$858K-\$1.7M in annual savings from 10%-20% portfolio-wide energy reductions.

Customer Purchase Prediction & Conversion Analytics

- Modeled purchase propensity across 12,205 cleaned sessions with Gradient Boosting, achieving 0.9377 ROC-AUC and 75.13% recall at a tuned 0.70 threshold; interpreted conversion drivers for campaign targeting and remarketing.

Breast Cancer Classification & Model Evaluation

- Compared logistic regression, tree ensembles, XGBoost, and neural networks on 569 records and 30 tumor features, achieving 98.25% test accuracy and 95.24% malignant-case recall with leakage-safe preprocessing and validation diagnostics.

Regression & Unsupervised Learning

- Built a TensorFlow/Keras housing regressor with 0.961 R-squared and approximately \$11.6K MAE, then implemented K-means from scratch and Gaussian anomaly detection to identify three traffic clusters and three energy-grid anomalies.

Visual Analytics & Business Intelligence

- Published three interactive Tableau dashboards spanning airline operations, workforce analytics, and location intelligence, using 6+ visual encodings, calculated fields, filters, parameters, tooltips, and dashboard-level KPI design.

EDUCATION & CERTIFICATIONS

Master of Science, Data Analytics (Big Data)

Completed Dec 2025

Canisius University, Buffalo, NY

Bachelor of Science, Economics

University of Abuja, Nigeria

Microsoft Certifications: Azure Fundamentals | Power BI Data Analyst Associate | Machine Learning Operations Engineer Associate

HONORS & SELECTED PRESENTATIONS

Honors: 1st Place, Data Masters Challenge (2025) | Koessler Summer Fellowship (2025) | Dean's List Excellence Award

Presentations: "Using AI to Address Cataloging Backlogs" (CCADE Fall 2025 Research Seminar) | "From Probability to Protection: Poisson Inequalities for Machine Learning in Cybersecurity" (IMPACT Faculty Celebration, 2025)